Processing and Interpreting Spectral Data Sets

I - Multispectral Image Analysis - THEMIS

During this section of the lab you will work through how to process and interpret multispectral data from the Mars Odyssey Thermal Emission Imaging System (THEMIS).

We're going to start with a pre-selected image to demonstrate some techniques. Open Matlab and load the data file 'lab_emissivity_THEMIS.mat'. This includes a multispectral image (I02207005) taken by the THEMIS instrument, onboard the Mars Odyssey spacecraft.

1. Introduction and Background

The Mars Odyssey Thermal Emission Imaging System (THEMIS) consists of a 5-band visible imager with 18 m/pixel spatial resolution, covering a wavelength range of 0.42 to 0.86 microns, and a 10-band infrared imager (100 m/pixel) covering 9 unique spectral bands between 6.8 and 14.9 microns [*Christensen et al.*, 2004]. (Bands 1 and 2 both cover the same wavelength range). In this lab we will use data from the infrared subsystem, over the region Nili Patera, a caldera of Syrtis Major. In class you learned that minerals exhibit diagnostic spectral features at infrared wavelengths. Here will use that information to interpret thermal infrared data from Mars.

THEMIS data are available from http://themis.asu.edu, in the form of calibrated radiance and brightness temperature. Note that THEMIS data products archived in the Planetary Data System have not been geometrically projected or band registered. Calibrated radiance images contain both surface and atmospheric contributions. The radiance measured by THEMIS, L(λ), is given by:

$$\mathbf{L(\lambda) = \varepsilon(\lambda)B(T_{surf}, \lambda)e^{-\tau_0 \lambda / \mu} + \int B(T(p), \lambda)e^{-\tau(\lambda, p)/\mu} d\tau} \quad (1)$$

Where $\varepsilon(\lambda)$ is the surface emissivity, B[T_{surf}, λ] is blackbody radiance as a function of surface temperature (T_{surf}), μ is the cosine of the emission angle, $\tau(\lambda,p)$ is the normal opacity profile as a function of pressure, and T(p) is the atmospheric

temperature profile. The integral is taken through the atmosphere from τ=0 (opacity at the spacecraft) to $\tau = \tau_0$ (opacity at the surface). The first term describes absorption of surface radiance ($\varepsilon(\lambda)B[T_{surf}, \lambda]$) by the atmosphere, while the second term describes radiation emitted by the atmosphere

[*Bandfield et al.*, 2004]. For this lab, the second term has already been accounted for in the radiance images, using the method described in *Bandfield et al.*, [2004].

2. Daytime radiance and introduction to the area of interest

Open the THEMIS radiance image for this lab by typing *bandview(radiance,wvl_radiance)*. This image has already been band registered for you using the software package ISIS. You can use the left and right arrows to browse the bands in this multispectral image cube, and q to quit. **Q1: What are we seeing in this daytime infrared image?**

Q2: How many surface units would you define based on what you see in the radiance image? Briefly describe them. (there's no right answer, by "surface unit", we just mean an area that has broadly similar characteristics - just use morphology and temperature differences to use your own judgement)

3. Deriving surface temperature and emissivity from radiance

Variations in surface radiance ($\varepsilon(\lambda)B[T_{surf},\lambda]$) from one region to the next can be due to either surface temperature, emissivity variations (likely composition), or both (eq. 1). To extract compositional information from a radiance cube, you must first perform a temperature-emissivity separation. For each radiance spectrum, there are n data points (radiance value at each channel n), but n+1 unknowns (emissivity at n, and T_{surf}). Thus either an emissivity at one channel must be assumed, or a temperature must be assumed to solve the equation [*e.g., Kahle and Alley*, 1992]. In this lab, we have used the Emissivity Normalization technique to separate temperature and emissivity [*Gillespie*, 1985; *Realmuto*, 1990]. For each pixel, a brightness temperature is calculated for each band using an emissivity value of 1.0. The maximum brightness temperature found for that spectrum is assumed to be the surface temperature (T_{surf}), and then T_{surf} is used to solve for emissivity for the other channels. In doing this, the emissivity values will be normalized to 1.0 and to the band with the maximum brightness temperature [*Christensen*, 1998]. Note

that surface emissivity is likely not equal to 1 across a full THEMIS band (~1 μm wide). Some researchers use a lower emissivity value (usually between 0.95 or 0.99), especially if a priori knowledge about the surface materials is available [*Kahle and Alley*, 1992]. In any case, it is important to remember that the derived emissivity values are relative to that of the channel set to 1.0; absolute values may be lower.

Q3: In what band do you see the highest radiance, and what temperature does this equate to? To estimate surface temperature more precisely, use *calc_temp(radiance(i,j,:),wvl_radiance(:))* to do a best fit of a pixel (i,j) of your choosing to the Planck function. As you can see, the fit plotted is not a particularly good fit to a Blackbody function. Let's instead use only bands 3:9 with *calc_temp(radiance(i,j,3:9),wvl_radiance(3:9))* – this should look a little better, although it's still not a perfect fit due to non-unity emissivity. How different is the result from your initial estimation based on the band with highest radiance? This blackbody function will have to be removed to look at emissivity variations due to mineral absorptions.

We have converted the radiance image to emissivity ahead of time using the ENVI software package. We have also removed bands 1,2, and 10 to focus on those bands which are most useful for examining surface variations. Check this out as before with *bandview(emiss,wvl_emiss)* Now, examine Band 5 emissivity. **Q4: How does this emissivity image look different than the radiance image? Why?**

Let's look at band combinations to get an idea of the variability. Use *imagergb(emiss,r,g,b)* where r, g, and b are three of the 7 available bands. Try looking at emissivity bands 8,7,5 or 9,6,4 (remember you will need to account for the fact that we have discarded bands 1 and 2) and plot them as red, green, and blue. **Q5: How does the image appear? Can you identify differing spectral units on the surface?** (Note: Spectral variations are often due to compositional variations, but not always, thus until full analysis is completed, we will call them spectral units). The 8-7-5 and 9-6-4 band combinations were pre-selected for you simply because they provide broad spectral coverage, but other band combinations may reveal different information.

II - Mineralogical Composition – TES

During this section of the lab you will work through how to process and interpret hyperspectral data from the Mars Global Surveyor Thermal Emission Spectrometer (TES).

1. Introduction and Background

The Thermal Emission Spectrometer (TES), one of the instruments on the Mars Global Surveyor (MGS) spacecraft, has lower spatial resolution than THEMIS, but many more spectral bands [*Christensen et al.*, 1992]. This allows for extraction of much more mineralogic detail. When used together, the TES and THEMIS data are a powerful compositional mapping tool. TES spectra from two distinct units identified in the THEMIS image have been provided for you. View the location of these units with *TES_roi*.

The TES spectra from each unit are from MGS mapping orbit 1927. These spectra contain contributions from the surface and atmosphere. Here you will perform the atmospheric correction using the method of *Smith et al.,* [2000]. Using a library of atmospheric spectral end-members derived by *Bandfield et al.* [2000] and mineral spectra, you will perform a linear least squares fit to the TES spectra.

To analyze the spectra, we will continue to use MATLAB. First, we need to read in the data files. In matlab, type

specinit

to read in the data for units 1 and 2, as well as the library of laboratory mineral spectra, and the library of atmospheric endmembers. Each of the items read in is a structure, with several variables each holding information about the data. Type the name of one of the variables (lib, atm, unit1, unit2) to see what's inside. Take a look at the unit1 spectrum with the command:

plotwb2(squeeze(unit1.xaxis),squeeze(unit1.data(1,:,:)))

This is a plot of emissivity (the ratio of the observed energy radiated by a material to that of a blackbody at the same temperature) vs. wavenumber (cm^{-1}). You should see that the spectrum is dominated by an absorption feature at ~700 wavenumbers.

Unfortunately, this is due to atmospheric CO_2 absorption, and only gets in the way of determining the surface mineralogy. There are also some areas at the ends of the spectrum that aren't considered for mineralogical analysis. In order to proceed, we'll need to crop out these areas of the spectrum with the commands:

unit1.data=cat(3,unit1.data(:,:,9:35),unit1.data(:,:,65:110));
unit2.data=cat(3,unit2.data(:,:,9:35),unit2.data(:,:,65:110));

The deconvolution code we will use will search for a linear combination of the library and atmospheric spectra that best fits the measured spectrum. Take a look at the mineral endmembers we'll be considering here, by typing:

lib.label

Here you should see a bunch of mineral names (Quartz, Microcline, etc., followed by various letters and numbers. The letters and numbers represent the catalog from which that mineral sample was acquired (e.g. WAR=Wards's).

2. Fitting a spectrum, the hard way

To perform the deconvolution on these two spectra, we'll use a function written by Oded Aharonson and Deanne Rogers, called decon. Type:

u1=decon(unit1.data,lib,[200 1305],[],atm);
u2=decon(unit2.data,lib,[200 1305],[],atm);

A bunch of text will scroll by showing the results of the deconvolution. We'll end up with the variables u1 and u2, which contain various results their respective structures. Before we actually look at the results, let's try to get a more intuitive feel for the fitting process. At the Matlab prompt, type:

specfit(u1,lib,atm);

This should bring up a plot window with a bunch of slider bars. The point of this routine is to give you a sense for how the automatic deconvolution works. The slider bars add different amounts of a given mineral to your fitted spectrum, while the plot shows the total sum of all the components you select, along with the current rms error between your fit and the measured spectrum. Both the raw (measured) and atmosphere-removed spectra are plotted. If your atmosphere component is set to

zero, the rms is calculated for the atmosphere removed spectra. Once you start adding atmosphere, the rms switches over to the measured spectrum. Try and find the best fit for the atmospherically corrected spectrum. **Q6: What is the lowest rms error you can find?** (If you can somehow beat the least squares value of .0019, you get an A!!) **Q7: What are the best fit values you find for each mineral group?**

Now, looking at the individual mineral group spectra with the slider bars, you should notice that each spectrum is dominated by one main absorption feature. For many of the mineral groups, this occurs at 10-11 microns, but for others, it occurs at longer wavelengths. **Q8: Try to guess what transitions might be (generally) responsible for these features**. (Hint: these energies are too low for electronic transitions, so we're primarily looking at vibrational modes).

3. Fitting a spectrum, the easy way

Okay, now let's look at the best fit model of the spectrum. At the matlab prompt, type:

specview2(u1,lib,atm);

You should see another window pop up, with a plot and some checkboxes for various the mineral groups. These checkboxes let you turn on and off the various mineral components (as determined by the decon function). **Q9: Which are the most important minerals to the best fit spectrum? How does this compare with the fit you found by hand? Q10: What are the fewest number of components which provide a reasonably good fit to the spectrum?** (no right answer for this one...)

Okay, now let's compare the unit1 and unit2 (which appeared different to THEMIS) spectra, to see how they appear to TES. We'll look at the atmosphere removed spectra, since we're interested in mineralogical differences. Type:

plotwb2(u1.xaxis,u1.rematm(:,1)); hold on;
plotwb2(u2.xaxis,u2.rematm(:,1),'r'); hold off

Unit1 will be blue, while unit2 will be red. You should see that there are some clear differences. Now, let's open up the fitted solutions. Type:

[u1.groupcoeff(:,1) u2.groupcoeff(:,1)]

This will bring up the coefficients for each of mineral groups, for unit1 (left column) and unit2 (right column). **Q11: What are the major differences in mineral abundances?** You can open up u1.groupnames if you need to see the names of the groups again. Now let's look at the standard deviations of the derived coefficients, to determine the significance of some of the results. Type:

[u1.groupcoeff(:,1) u1.groupstd(:,1) u2.groupcoeff(:,1) u2.groupstd(:,1)]

Q12: Now, looking at the standard error magnitudes, are there any mineral detections that are not statistically significant (at the 1-sigma level)? Q13: Is the difference in say, carbonate between the two units significant? What about the difference in high-silica phases?

III - Interpreting the geology

Q14: Using the compositional, morphologic, and stratigraphic relationships present in the various datasets you have analyzed, what can you say about the geologic history of this area? Obviously, you can't know every single event and process that has occurred, but try to say what you can. Additional context images from MOC and THEMIS are included in the lab files provided.

Now that you're all finished, we can tell you that a similar analysis of this region was presented in *Christensen et al.* [2005]. Check it out if you are interested!

That's it, good luck!